

An Analytical Study of Consumer Financial Complaint Resolution

Outcomes



Mahitha Kalinathabotla
University at Albany

Dr. Abhishek Ghosh
Professor
University at Albany
1400 Washington Ave
Albany NY 12222

Written Analysis
15 February 2026
Word count: 3484

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Abstract

This study explores consumer financial complaints using data from the Consumer Financial Protection Bureau (CFPB) to better understand how complaint characteristics and company response behavior influence complaint resolution outcomes and response times. The CFPB Consumer Complaint Database contains millions of complaints related to financial products such as credit cards, mortgages, and credit reporting services. Using a dataset of approximately 5.6 million resolved complaints, this study uses descriptive statistics and exploratory data analysis to identify patterns in how companies respond to complaints and whether consumers receive relief. Preliminary results show that most companies respond to complaints on the same day they are received. However, response times vary widely, with some complaints taking much longer to address. Complaints involving more complex financial products also tend to require longer response times.

Introduction

Consumer financial complaints offer useful insight into how individuals experience financial products and services such as credit cards, loans, and mortgages. These complaints often arise when consumers encounter issues related to billing errors, delays in service, or unclear communication from financial institutions. In the United States, the Consumer Financial Protection Bureau (CFPB) collects and publishes these complaints through its Consumer Complaint Database.

The CFPB Consumer Complaint Database contains detailed information for each complaint, including the financial product involved, the nature of the issue, the company associated with the complaint, the company's response, response timelines, and the final resolution outcome. Because

the dataset is publicly available, large in scale, and well structured, it provides a strong foundation for analyzing patterns in complaint handling.

Studying how consumer financial complaints are resolved is important for several reasons. From the consumer perspective, timely and fair complaint resolution helps build trust in financial institutions and improves overall customer satisfaction. Delays or unfavorable outcomes may reduce consumer confidence and negatively impact their experience with financial services.

From a business perspective, financial institutions receive many complaints each year, and inefficient complaint handling can result in reputational damage, increased regulatory scrutiny, and higher operational costs. By understanding which complaint characteristics and company response behaviors influence resolution outcomes and response time, companies can improve their complaint management processes and enhance service quality.

From a regulatory standpoint, organizations such as the CFPB use complaint data to monitor company behavior and identify potential issues in financial markets. Applying business analytics and predictive modeling techniques to this data can provide valuable insights that support consumer protection efforts and improve transparency in financial services.

Based on this context, this study aims to examine the following research question:

To what extent do the characteristics of consumer financial complaints and the way companies respond to them influence whether complaints are resolved in favor of the consumer? Additionally, how do these factors affect the time required to resolve consumer financial complaints?

Literature Review

Consumer complaint systems play an important role in helping researchers and regulators understand how financial institutions interact with their customers. In the United States, the Consumer Financial Protection Bureau (CFPB) manages a publicly available database that contains detailed records of consumer complaints related to various financial products and services. This database has been used extensively by researchers and policymakers to analyze trends in consumer dissatisfaction and evaluate company response behavior. The structured nature of this complaint-level data makes it possible to examine not only the number of complaints filed but also how financial institutions respond and how those complaints are eventually resolved (Consumer Financial Protection Bureau, 2026).

Early research examining CFPB complaint data has shown that complaint outcomes and company responses are not the same across all financial products or institutions. Ayres, Lingwall, and Steinway (2014) analyzed complaint records and found noticeable differences in how companies handled consumer concerns. Their findings indicated that some financial institutions demonstrated more consistent and effective response practices, while others showed less responsiveness. This study highlighted that complaint-level data can be used to assess company behavior and evaluate how financial institutions manage customer issues. It also suggested that both the nature of the complaint and the actions taken by the company may influence how complaints are ultimately resolved.

Beyond complaint characteristics, company response behavior has also been identified as an important factor affecting resolution outcomes and organizational accountability. Dou and Roh (2019) found that making complaint information publicly accessible can motivate financial

institutions to improve their response practices. When complaint data is transparent, companies may be more likely to respond promptly and resolve issues efficiently in order to maintain their reputation and avoid regulatory concerns. These findings emphasize the importance of response timeliness and quality in determining how effectively complaints are handled and resolved.

Recent research has also explored the use of data analytics and machine learning to analyze consumer complaint data and predict resolution outcomes. Vaishnav et al. (2024) demonstrated that structured complaint data, including information about financial products, issue types, and company responses, can be used to develop predictive models that identify patterns in complaint handling. In a similar study, Oyewola et al. (2023) applied machine learning techniques to CFPB complaint data and showed that these models can successfully classify and analyze complaint records. Their work highlights the potential of predictive analytics to support more efficient complaint management and improve understanding of complaint resolution processes.

Although prior studies have shown that complaint data and predictive modeling can provide valuable insights, there is still limited research examining how complaint characteristics and company response behavior together influence both resolution outcomes and the time required to resolve complaints. This study builds on existing literature by analyzing CFPB complaint data to investigate the relationship between complaint features, company responses, and resolution efficiency. By applying business analytics methods, this research aims to identify key factors associated with complaint resolution outcomes and contribute to a better understanding of complaint handling in the financial services industry. These studies provide a strong foundation for examining how complaint characteristics and company response behavior influence complaint

resolution outcomes and resolution time. Nonetheless, only a limited number of studies have looked at both resolution outcome and resolution time at the same time.

Hypotheses

Based on the research question and findings from prior studies, this research proposes several hypotheses related to complaint characteristics, company response behavior, resolution outcomes, and resolution time.

Hypothesis 1 (H1): Complaints regarding mortgages and credit reporting are less likely to lead to financial compensation for consumers than those associated with credit cards and retail banking services.

Complaints regarding mortgages and credit reporting often encompass more intricate financial and legal issues, potentially lowering the chances of receiving monetary compensation relative to complaints related to credit cards and retail banking services.

Hypothesis 2 (H2): Complaints in which companies respond promptly are more likely to be settled in favor of the consumer than those with slow company replies.

Faster initial forwarding of complaints to companies may improve coordination and processing efficiency, which can indirectly influence resolution outcomes.

Hypothesis 3 (H3): The typical time for resolution is greater for complaints related to mortgages and credit reporting than for those concerning credit cards and retail banking.

Complaints about mortgages and credit reporting usually require more in-depth investigation and paperwork, potentially prolonging the average resolution time relative to complaints about simpler financial products.

Hypothesis 4 (H4): Complaints that receive delayed responses from the company take longer on average to resolve than those with prompt responses.

Delayed replies from the company can hinder the complaint resolution process, leading to an extended overall resolution duration.

Methodology

This research utilizes the Consumer Financial Protection Bureau (CFPB) Consumer Complaint Database, a publicly accessible dataset that logs complaints filed by consumers about financial products and services. The dataset includes details about the financial product in question, the consumer's reported issue, the company that received the complaint, the company's response actions, and the final resolution result. This analysis involves a dataset of about 5.63 million resolved complaint entries.

The data was processed and analyzed using Python along with the pandas, seaborn, and matplotlib libraries. The initial preprocessing involved cleaning the dataset, standardizing date variables into a consistent datetime format, and eliminating entries with invalid or negative response times. This guaranteed that the response time variable correctly indicated the days elapsed from when a complaint was received to when it was forwarded to the company.

It is important to note that response time and time-to-resolution are distinct measures. In this study, response time is defined as the number of days between when a complaint is received and when it is forwarded to the company. It does not refer to the total time required for full complaint resolution.

Two primary variables were created for the analysis. The initial variable, response time, was determined by subtracting the complaint submission date from the date the complaint was sent to the company. This variable indicates the speed at which companies address consumer complaints.

The second variable, consumer relief, was established as a binary indicator indicating whether a complaint led to financial or non-financial assistance for the consumer. Complaints that resulted in relief were assigned a code of 1, whereas those resolved without relief were given a code of 0. Techniques for exploratory data analysis were employed to investigate trends in complaint resolution and response duration. Descriptive statistics were computed to encapsulate the distribution of response times, while visual representations like histograms and bar graphs were employed to uncover associations among complaint features, response rates, and resolution results. Moreover, response times were divided into various categories indicating distinct levels of speed, such as same-day replies, brief delays, moderate delays, and extended response durations. These categories were utilized to examine if quicker company reactions correlate with an increased probability of consumer satisfaction.

This approach enables a structured analysis of how complaint attributes and corporate reaction patterns affect both customer satisfaction results and the duration needed to address complaints.

Descriptive and Exploratory Analysis

This section showcases descriptive statistics and exploratory visualizations of the Consumer Complaint dataset from the CFPB. The goal is to outline the main variables utilized in the analysis and recognize initial trends associated with complaint attributes, corporate response actions, consumer satisfaction results, and response duration. These preliminary results offer early support for the suggested hypotheses and guide the forthcoming statistical evaluation.

Summary Statistics

Table 1. Summary Statistics

Variable	Count	Mean	Median	Min	Max	Std Dev
Response Time	5,631,692	0.55	0	0	1962	4.92

The dataset contains approximately **5.63 million resolved complaints**. The average company response time is **0.55 days**, while the median response time is **0 days**, indicating that most companies respond to complaints on the same day they are received. However, the maximum response time reaches **1,962 days**, suggesting the presence of extreme outliers and a highly right-skewed distribution.

In terms of resolution outcomes, complaints were categorized into two groups: those resulting in consumer relief and those resolved without relief. Around 39% of grievances led to various types of financial or non-financial remedies, whereas most were settled without any direct compensation to consumers.

Table 2. Distribution of Company Response Time

Statistic	Response Time (Days)
Count	5,631,692
Mean	0.55
Standard Deviation	4.92
Minimum	0
25th Percentile	0

Median(50th Percentile)	0
75th Percentile	0
Maximum	1962

Table 2 provides further descriptive statistics regarding the response time of companies. The data shows that the response times are primarily clustered around lower values, with the median response time being zero days. This indicates that numerous companies address consumer complaints on the day they are submitted. Yet, the highest response time reaches 1,962 days, showing the existence of significant outliers and a lengthy right tail in the distribution. These findings indicate that although responses are generally quick, a few instances encounter significantly longer wait times.

Figure 1. Distribution of Response Time

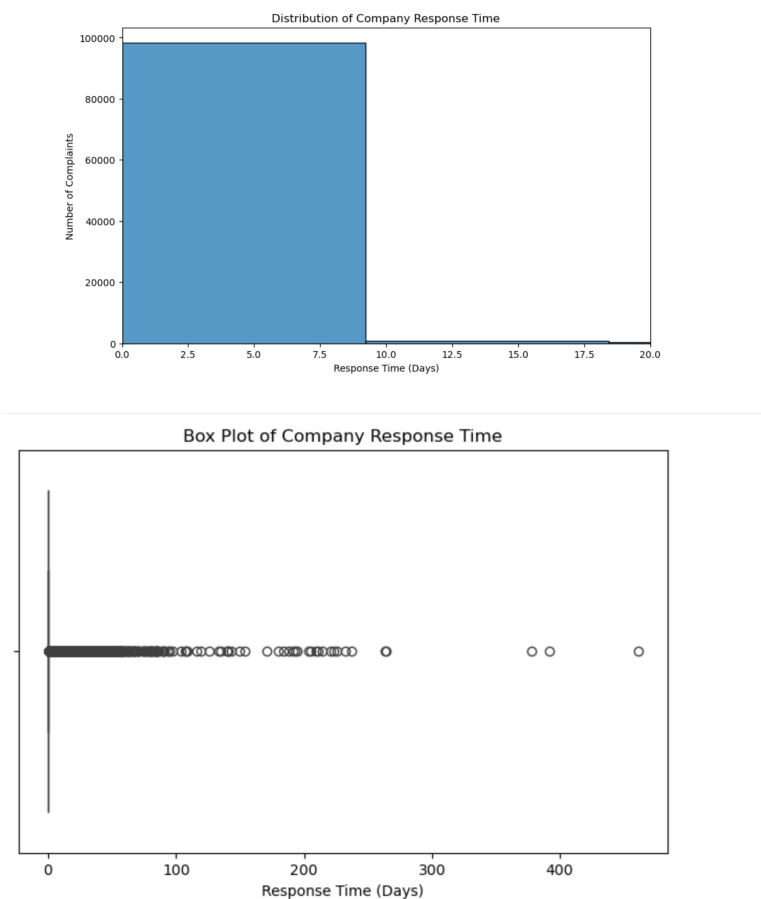


Figure 1 shows the allocation of response times from companies in relation to complaints. The distribution is heavily right skewed, with most complaints getting a response in under one day. Nonetheless, a limited number of instances encounter significantly extended response periods, resulting in an elongated right tail.

Figure 2. Average Company Response Time by Financial Product

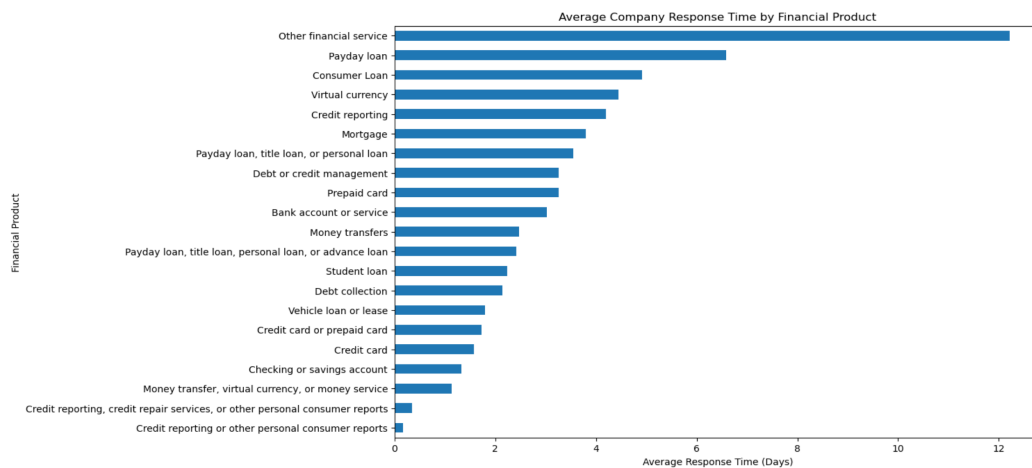


Figure 2 depicts the average response time for various categories of financial products. Although many financial products demonstrate relatively short response times, some categories show slightly longer averages. This suggests that complaints involving more complex financial products may require additional review before resolution.

Figure 3. Response Time by Consumer Relief Outcome

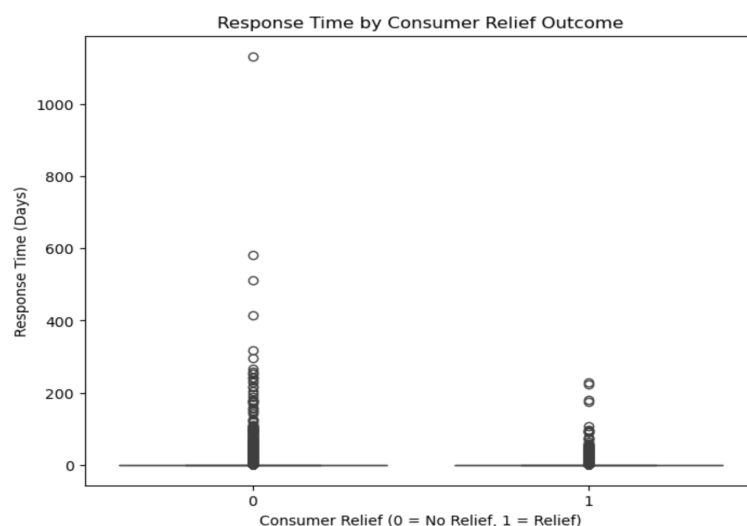


Figure 3 contrasts the response times for complaints that led to consumer relief with those that did not result in such relief. Complaints resulting in consumer relief generally show somewhat longer

average response times, indicating that cases needing compensation or corrective measures might necessitate further examination. This suggests that cases involving financial compensation may require additional verification or investigation before resolution.

Figure 4. Consumer Relief Rate by Response Time

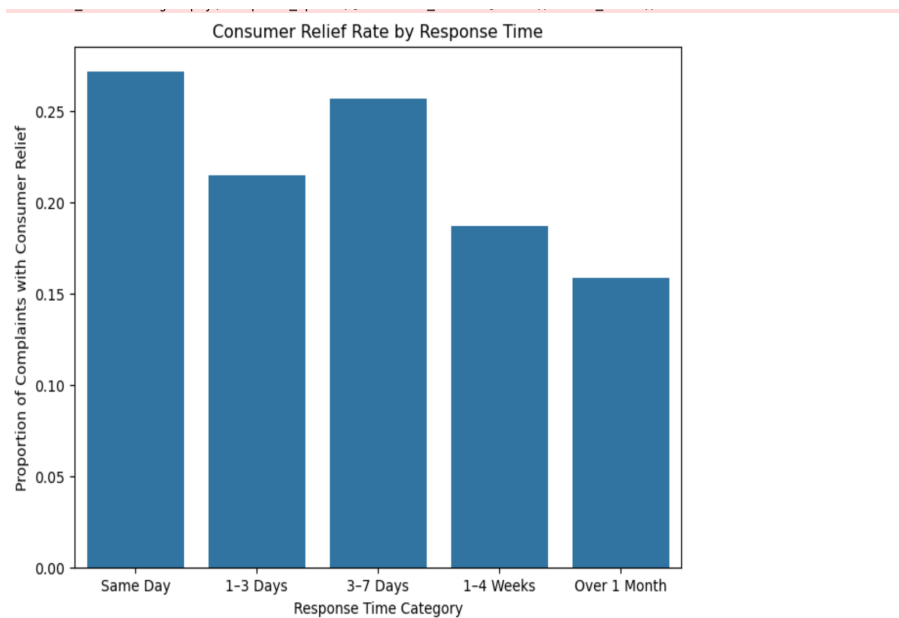


Figure 4 illustrates the percentage of complaints that led to consumer relief among various response time. The findings indicate that the speed of response could be linked to the chances of consumer satisfaction, though the variations among categories are minor.

Figure 5. Average Response Time by Financial Product and Consumer Relief Outcome

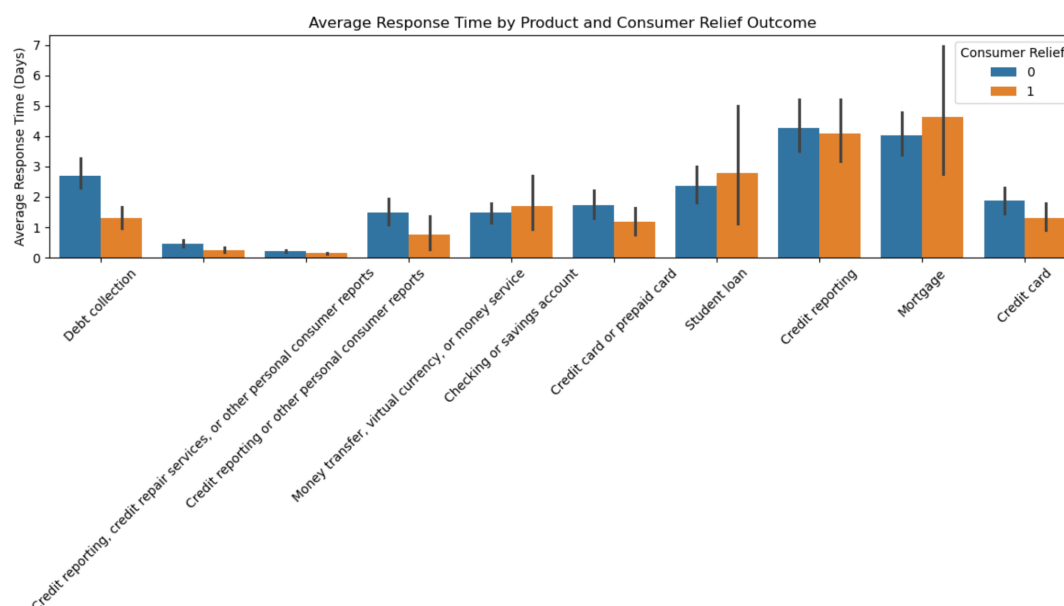


Figure 5 depicts the relationship between categories of financial products and the results of consumer relief. In various product categories, complaints leading to consumer relief generally take a bit longer to resolve compared to those settled without relief, suggesting that more complicated cases might need further examination.

Conclusion

This research examined trends in resolving consumer financial complaints utilizing information from the CFPB Consumer Complaint Database. The exploratory analysis suggests that the behavior of company responses and the traits of financial products could affect resolution outcomes and response duration. Although most complaints get replies within a day, a few cases face much longer wait times, resulting in a highly skewed distribution of response times. The results indicate that grievances related to more intricate financial products may need extended review times and are more prone to lead to consumer compensation. These initial findings offer

early backing for the suggested hypotheses and establish a basis for upcoming statistical modeling to rigorously evaluate these connections.

Methods and Preliminary Results

Building on the exploratory analysis presented earlier, several patterns emerged between complaint characteristics, company response behavior, and complaint resolution outcomes. While these initial observations provide useful insights, more rigorous statistical methods are needed to formally test the hypotheses and better understand these relationships.

To address this, the study applies regression-based modeling techniques that match the type of outcome being analyzed. Logistic regression is used to examine consumer relief, since it is a binary outcome (1 = relief, 0 = no relief). This approach allows us to estimate the likelihood of a complaint resulting in relief while accounting for multiple influencing factors.

Response time, on the other hand, is highly right-skewed, so alternative approaches such as categorization or transformation are considered instead of relying solely on standard linear regression. This ensures that the analysis better reflects the underlying distribution of the data.

To improve the accuracy of the estimates and reduce omitted variable bias, a multivariable framework is used. In addition to response time, the model includes financial product categories as control variables, allowing the analysis to capture differences across types of complaints.

To further examine whether the effect of response time varies across different types of complaints, interaction terms between response time and financial product categories were incorporated into the logistic regression model.

The results indicate that the interaction between response time and mortgage-related complaints is positive and statistically significant (coefficient = 0.0207, $p < 0.001$). While the baseline effect of response time remains negative (coefficient = -0.0185, $p < 0.001$), the positive interaction term suggests that this negative effect is attenuated for mortgage complaints.

In addition to mortgage interactions, interaction terms between response time and all product categories were included in the model. This allows the effect of response time to vary across different financial products rather than focusing on a single category.

Specifically, although longer response times are generally associated with a lower likelihood of consumer relief, this relationship is weaker for mortgage-related complaints. This may reflect the more complex nature of mortgage cases, which often require additional time for review and processing.

Overall, these findings highlight that the impact of response time on complaint outcomes is not uniform across product categories. Instead, the relationship varies depending on the type of financial product, providing additional support for the importance of accounting for interaction effects in the analysis.

The logistic regression model is specified as follows:

$$\text{logit}(P(\text{Consumer Relief} = 1)) = \beta_0 + \beta_1(\text{Response Time}) + \beta_2(\text{Product Categories})$$

where Consumer Relief indicates whether a complaint resulted in relief, Response Time measures the number of days between when the complaint was received and when it was sent to

the company, and Product Categories are included as dummy variables to capture variation across financial products.

Additionally, interaction terms between response time and product categories are included to capture heterogeneous effects across complaint types.

To address limitations of the initial analysis, a refined multivariable logistic regression model was estimated by incorporating these additional variables.

The results show that response time has a statistically significant negative effect on the likelihood of consumer relief (coefficient = -0.0140, $p < 0.001$). This means that as response time increases, the probability of a complaint being resolved in favor of the consumer decreases. In simpler terms, faster responses are associated with better outcomes for consumers.

Compared to the earlier bivariate model, the magnitude of the response time coefficient is smaller. This suggests that part of the initial effect was influenced by differences across complaint types, highlighting the importance of including relevant control variables in the model.

Table 3 presents the results of the refined multivariable logistic regression model.

Variable	Coefficient	Std. Error	z-value	p-value
Intercept	-0.7757	0.088	-8.781	<0.001
Response Time	-0.0140	0.002	-6.230	<0.001
Credit Card	0.2531	0.099	2.555	0.011
Mortgage	-1.4515	0.107	-13.574	<0.001
Debt Collection	-0.6064	0.093	-6.520	<0.001

Note: Additional product category coefficients are included in the model but omitted from the table for brevity.

In addition to response time, the results show clear differences across financial product categories. For instance, complaints related to mortgages and certain loan products are less likely to result in consumer relief, while complaints involving credit cards tend to have more favorable outcomes. This highlights the importance of accounting for complaint characteristics when analyzing resolution outcomes.

The inclusion of additional variables improves the model's ability to explain variation in complaint outcomes compared to the initial specification. The model is also statistically significant overall ($p < 0.001$), indicating that the included variables meaningfully contribute to explaining the results.

The assumptions of logistic regression were also considered. The dependent variable meets the binary requirement of the model. *Dummy variables were constructed with Credit Reporting used as the baseline reference category. Therefore, all coefficients represent effects relative to complaints involving credit reporting.* The large sample size further strengthens the reliability of the estimates.

Overall, these findings provide strong support for the hypothesis that timely company responses are associated with more favorable complaint resolution outcomes. At the same time, the results emphasize that complaint characteristics, particularly product type, play a significant role in determining outcomes.

To provide a more complete understanding, interaction effects between response time and product categories are also explored. This allows the analysis to capture how the relationship between response behavior and outcomes may differ across complaint types.

For hypotheses related to response time (H3 and H4), standard linear regression is not appropriate due to the count-based and highly right-skewed nature of the response time variable. Therefore, a generalized linear modeling (GLM) framework is more suitable.

In particular, a negative binomial regression model is appropriate, as it accounts for overdispersion in count data and better captures the distribution of response times. This approach will be implemented in the next stage of the analysis. This extension will allow for a more accurate examination of factors influencing response time.

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